KAIZEN MATHS

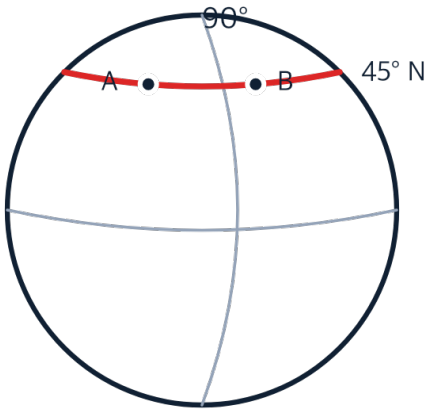
Lesson starter

Earth Geometry

Parallels and Routes

Name _____

SA1



Schematic globe; points lie on the shown latitude or meridian

A parallel of latitude is 45° N. Taking the Earth radius as 6400 km, find the radius of this small circle.

GEO-F170-L2 · SA

Show your working

KAIZEN MATHS

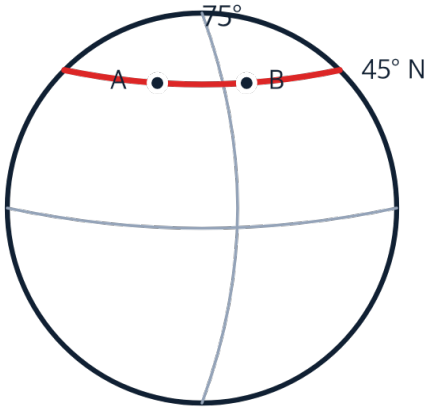
Lesson starter

Earth Geometry

Parallels and Routes

Name _____

SB1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on latitude 45° N and their longitudes differ by 75°. Taking the Earth radius as 6400 km, find the distance from A to B along this parallel.

GEO-F170-L2 · SB

Show your working

KAIZEN MATHS

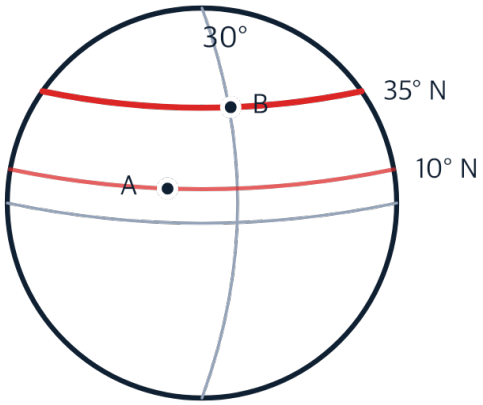
Lesson starter

Earth Geometry

Parallels and Routes

Name _____

SC1



Schematic globe; points lie on the shown latitude or meridian

A ship travels from A at 10° N to B at 35° N by first travelling north-south along a meridian, then travelling 30° of longitude along latitude 35° N. Taking R = 6400 km, find the total distance travelled.

GEO-F170-L2 · SC

Show your working

KAIZEN MATHS

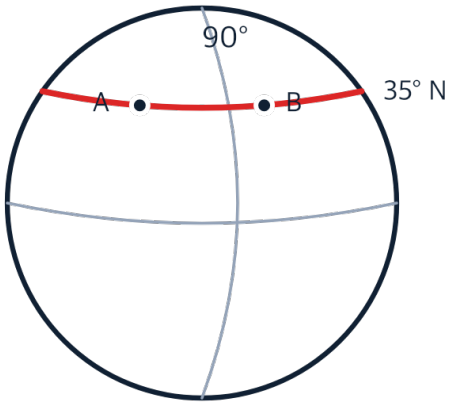
Lesson starter

Earth Geometry

Parallels and Routes

Name _____

SD1



Schematic globe; points lie on the shown latitude or meridian

A parallel of latitude is 35° N. Taking the Earth radius as 6400 km, find the radius of this small circle.

KAIZEN MATHS

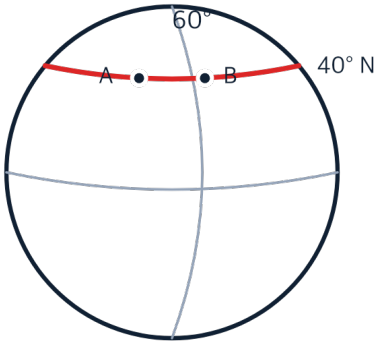
Exit ticket

Earth Geometry

Parallels and Routes

Name _____

EA1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on latitude 40° N and their longitudes differ by 60°. Taking the Earth radius as 6400 km, find the distance from A to B along this parallel.

GEO-F170-L2 · EA

Show your working

KAIZEN MATHS

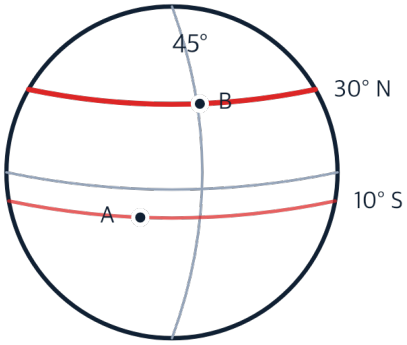
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Earth Geometry

Parallels and Routes

Name _____

EB1



Schematic globe; points lie on the shown latitude or meridian

A ship travels from A at 10° S to B at 30° N by first travelling north-south along a meridian, then travelling 45° of longitude along latitude 30° N. Taking R = 6400 km, find the total distance travelled.

GEO-F170-L2 · EB

Show your working

KAIZEN MATHS

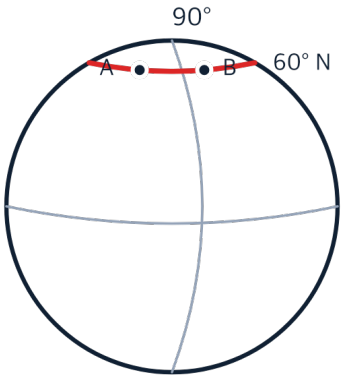
Exit ticket

Earth Geometry

Parallels and Routes

Name _____

EC1



Schematic globe; points lie on the shown latitude or meridian

A parallel of latitude is 60° N. Taking the Earth radius as 6400 km, find the radius of this small circle.

GEO-F170-L2 · EC

Show your working

KAIZEN MATHS

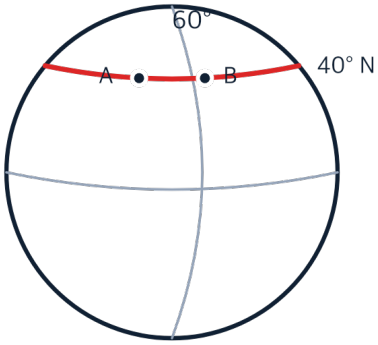
Exit ticket

Earth Geometry

Parallels and Routes

Name _____

EA1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on latitude 40° N and their longitudes differ by 60°. Taking the Earth radius as 6400 km, find the distance from A to B along this parallel.

GEO-F170-L2 · EA

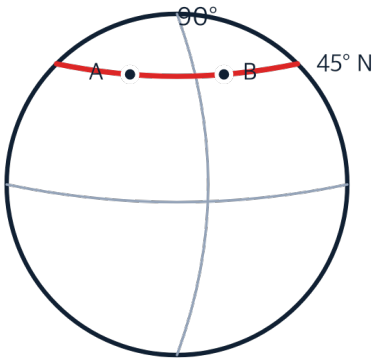
Show your working

Teacher answers & working

Earth Geometry · Parallels and Routes

GEO-F170-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Schematic globe; points lie on the shown latitude or meridian

A parallel of latitude is 45° N. Taking the Earth radius as 6400 km, find the radius of this small circle.

Answer:

4,525 km

Working:

Step 1: A parallel of latitude is a small circle.

Step 2: Its radius is R cos(latitude).

radius = 6400 cos(45°)

radius ≈ 4525.5 km

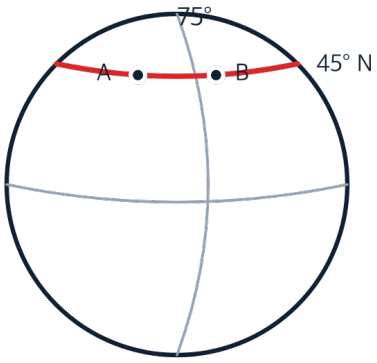
Final answer: 4,525 km

Teacher answers & working

Earth Geometry · Parallels and Routes

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SB1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on latitude 45° N and their longitudes differ by 75°. Taking the Earth radius as 6400 km, find the distance from A to B along this parallel.

Answer:

5,924 km

Working:

Step 1: The path is along a parallel, not a great circle.

Step 2: Find the radius of the parallel.

small-circle radius = R cos(45°)

small-circle radius = 6400 cos(45°) ≈ 4525.5 km

Step 3: Use arc length with the small-circle radius.

distance = (75 ÷ 360) × 2π × 4525.5

distance ≈ 5923.8 km

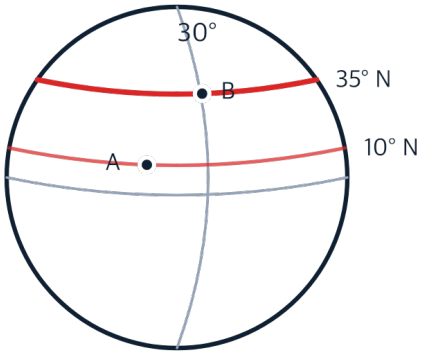
Final answer: 5,924 km

Teacher answers & working

Earth Geometry · Parallels and Routes

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SC1



Schematic globe; points lie on the shown latitude or meridian

A ship travels from A at 10° N to B at 35° N by first travelling north-south along a meridian, then travelling 30° of longitude along latitude 35° N. Taking R = 6400 km, find the total distance travelled.

Answer:

5,538 km

Working:

Step 1: Split the journey into two sections.

Section 1 is along a meridian, so use the full Earth radius.

latitude difference = |35 - 10| = 25°

meridian distance = (25 ÷ 360) × 2π × 6400 ≈ 2792.5 km

Step 2: Section 2 is along a parallel at 35° N.

parallel radius = 6400 cos(35°) ≈ 5242.6 km

parallel distance = (30 ÷ 360) × 2π × 5242.6 ≈ 2745.0 km

Step 3: Add the two distances.

total distance ≈ 2792.5 + 2745.0 = 5537.5 km

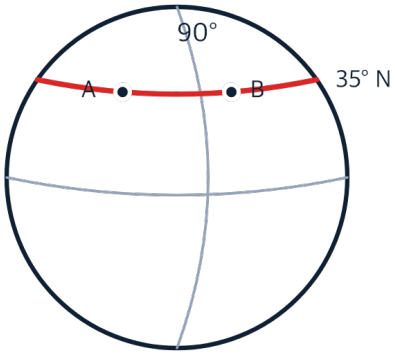
Final answer: 5,538 km

Teacher answers & working

Earth Geometry · Parallels and Routes

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SD1



Schematic globe; points lie on the shown latitude or meridian

A parallel of latitude is 35° N. Taking the Earth radius as 6400 km, find the radius of this small circle.

Answer:

5,243 km

Working:

Step 1: A parallel of latitude is a small circle.

Step 2: Its radius is R cos(latitude).

radius = 6400 cos(35°)

radius ≈ 5242.6 km

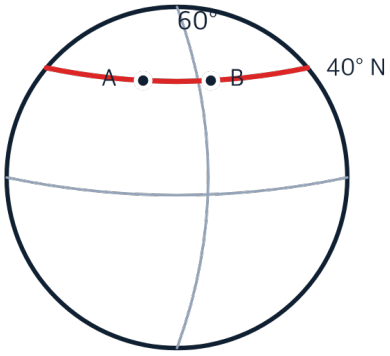
Final answer: 5,243 km

Teacher answers & working

Earth Geometry · Parallels and Routes

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EA1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on latitude 40° N and their longitudes differ by 60°. Taking the Earth radius as 6400 km, find the distance from A to B along this parallel.

Answer:

5,134 km

Working:

Step 1: The path is along a parallel, not a great circle.

Step 2: Find the radius of the parallel.

small-circle radius = R cos(40°)

small-circle radius = 6400 cos(40°) ≈ 4902.7 km

Step 3: Use arc length with the small-circle radius.

distance = (60 ÷ 360) × 2π × 4902.7

distance ≈ 5134.1 km

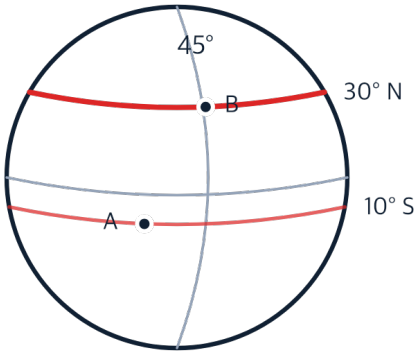
Final answer: 5,134 km

Teacher answers & working

Earth Geometry · Parallels and Routes

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EB1



Schematic globe; points lie on the shown latitude or meridian

A ship travels from A at 10° S to B at 30° N by first travelling north-south along a meridian, then travelling 45° of longitude along latitude 30° N. Taking R = 6400 km, find the total distance travelled.

Answer:

8,821 km

Working:

Step 1: Split the journey into two sections.

Section 1 is along a meridian, so use the full Earth radius.

latitude difference = |30 - -10| = 40°

meridian distance = (40 ÷ 360) × 2π × 6400 ≈ 4468.0 km

Step 2: Section 2 is along a parallel at 30° N.

parallel radius = 6400 cos(30°) ≈ 5542.6 km

parallel distance = (45 ÷ 360) × 2π × 5542.6 ≈ 4353.1 km

Step 3: Add the two distances.

total distance ≈ 4468.0 + 4353.1 = 8821.2 km

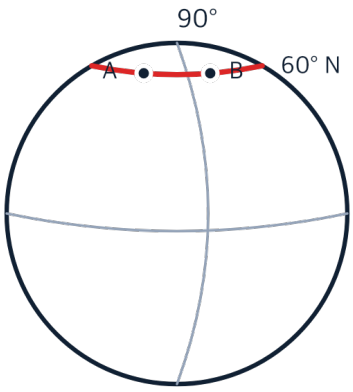
Final answer: 8,821 km

Teacher answers & working

Earth Geometry · Parallels and Routes

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EC1



Schematic globe; points lie on the shown latitude or meridian

A parallel of latitude is 60° N. Taking the Earth radius as 6400 km, find the radius of this small circle.

Answer:

3,200 km

Working:

Step 1: A parallel of latitude is a small circle.

Step 2: Its radius is R cos(latitude).

radius = 6400 cos(60°)

radius ≈ 3200.0 km

Final answer: 3,200 km